

EpiKG: End-to-End Epistemic Preservation in Clinical Knowledge Graphs for Assertion-Aware Retrieval-Augmented Generation

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Abstract

Clinical NLP detects assertion status—negation, uncertainty, family history—yet no KG-RAG system propagates these labels into retrieval-augmented generation. We call this the *epistemic propagation gap*. EpiKG closes it with a 7-value assertion schema on every edge, tri-temporal modeling, and intent-aware retrieval that dispatches question-type-specific graph traversals. On ClinicalBench ($n = 400$, MIMIC-IV, Claude Opus 4.6), intent-aware KG-RAG achieves 69% (+47 pp vs. LLM alone), improving all 9 categories. An ablation decomposition isolates two additive components: structured retrieval contributes +28 pp, epistemic annotations a further +19 pp—a roughly 60/40 split. Change detection exemplifies the distinction: retrieval without assertions reaches 17%, but adding assertion metadata lifts it to 100%. Brute-force long context collapses to 0% on historical and sequence despite a 200K token window. Cross-model validation on three models shows KG-RAG benefit is consistent with an inverse relationship to parametric medical knowledge, suggesting a knowledge-compensator effect. The category $\times$ condition interaction—not the aggregate—is the right evaluation lens for clinical KG-RAG.

1 Introduction

A physician documents “patient denies chest pain, sister had MI at 45, will consider statin if lipids remain elevated”—a single sentence encoding four epistemic states: a negated symptom, a family-history event, a hypothetical action, and an implicit present condition. Clinical NLP can detect these assertions accurately [1, 2], yet when a RAG system ingests the same note, assertion context is flattened: negation is lost, family attribution collapses onto the patient, and downstream reasoning may conclude the patient *has* chest pain and *had* an MI. We call this the *epistemic propagation gap*—the systematic loss of assertion status as clinical information moves from extraction through storage to retrieval.

The gap persists because no KG-RAG system, to our knowledge, propagates assertion labels beyond the extraction stage into graph-augmented retrieval. Clinical NLP pipelines (cTAKES, John Snow Labs) detect assertions, but do not carry them as edge attributes into knowledge graph construction or retrieval. OMOP (Observational Medical Outcomes Partnership) excludes negated conditions from `CONDITION_OCCURRENCE` [3]; FHIR (Fast Healthcare Interoperability Resources) provides `verificationStatus` for `Condition` resources only, covering a fraction of the clinical data model. Graph-augmented RAG systems—GraphRAG [4], GFM-RAG [5], KARE [6], Medical-Graph-RAG [7]—build KGs that discard the metadata distinguishing “patient has diabetes” from “rule out diabetes.” Two KG surveys [8, 9] do not identify assertion preservation as a concern. A parallel *temporal integration gap* exists: clinical events span three time dimensions (valid time, transaction time, NLP-asserted temporality), yet existing systems model at most two [10, 11]. Allen’s interval algebra [12] has been formalized as annotation ontologies [13] and modal operators [14], but never as first-class KG edge attributes that participate in retrieval.

We present EpiKG, a clinical KG system that closes both gaps by preserving epistemic status end-to-end from NLP extraction through OMOP mapping, fact construction, KG materialization, and graph-augmented retrieval. Our contributions:

1. **Epistemic preservation invariant.** We formalize the requirement that assertion status be recoverable at every downstream stage, give an information-theoretic bound on the loss from assertion-blind pipelines (Section 3.5, Appendix B), and realize the invariant with a seven-value schema extending i2b2 [1], carried from extraction through OMOP mapping, fact construction, KG materialization, and RAG serialization.
2. **Tri-temporal KG with Allen’s interval algebra.** Edges carry valid time, transaction time, and NLP-asserted temporality alongside nine temporal relations (seven from Allen’s algebra [12] plus CONCURRENT and UNKNOWN) as first-class attributes, enabling temporal reasoning over longitudinal records.
3. **Ablation with decomposition and bookend baselines.** We evaluate retrieval conditions—from C7 (deterministic KG, no LLM; 3.8%) through C4g (intent-aware KG-RAG; 69.0%, +47 pp vs. LLM alone)—on ClinicalBench and SliceBench. A new ablation step C3 (KG-RAG without assertions; 50.0%) decomposes the gain: structured retrieval contributes +28 pp, epistemic annotations +19 pp. Bookend baselines C6 (brute-force long context; 39.0%) and C7 bracket the design space. Intent-aware KG-RAG improves all 9 categories; neither more context nor structured data alone can substitute for LLM + epistemic KG-RAG.
4. **ClinicalBench, SliceBench, and cross-model validation.** Two benchmarks (400 and 144 questions) across 9 epistemic categories target the propagation gap, with hard longitudinal questions (change, cross-admission, temporal) as the primary endpoint. Cross-model validation on MedGemma 27B and GPT-OSS 20B shows KG-RAG benefit consistent with an inverse relationship to parametric medical knowledge; a two-physician blinded adjudication is preregistered (Appendix O).

The central research question is: *under what conditions does structured epistemic context help, hurt, or break even compared to unstructured retrieval or no retrieval at all?* The answer is decisive: intent-aware KG-RAG improves all 9 categories, with the largest gains on longitudinal reasoning that requires cross-admission synthesis. An ablation decomposition shows that structured retrieval provides the majority of the gain (+28 pp); epistemic annotations are critical for change detection, where assertion metadata alone accounts for an +83 pp jump. Brute-force long context collapses on historical and sequence categories despite a 200K token window. Cross-model validation shows the benefit is consistent with an inverse relationship to parametric medical knowledge, suggesting a knowledge-compensator effect. The magnitude and direction of the effect are both category- and model-specific, producing interactions that aggregate accuracy alone obscures (Section 5).

2 Related Work

We organize prior work along four axes and synthesize the gaps that EpiKG addresses (Table 7, Appendix).

Medical RAG systems. Graph-augmented retrieval has emerged as a leading paradigm for medical QA. GraphRAG [4] introduced community-based summarization; GFM-RAG [5] trained an 8M-parameter graph foundation model across 60 KGs (14M+ triples); KARE [6] adapted community retrieval to clinical decision support; DoctorRAG [15] emulates physician reasoning; MedRAG [16] constructs hierarchical diagnostic KGs; Medical-Graph-RAG [7] links documents via triple graphs. Concurrent work addresses EHR-specific retrieval: EHR-RAG [17] evaluates retrieval strategies over structured EHR tables, and MediGRAF [18] builds medical KGs via multi-agent extraction. Yet no published medical RAG system models assertion status or temporal context; GFM-RAG and KARE build population-level graphs rather than patient-level graphs from clinical text, and a recent survey [19] confirms no system incorporates epistemic metadata into the retrieval graph.

Clinical KG construction. Multi-LLM KG-RAG [20] uses multi-agent LLM prompting with schema-constrained extraction for oncology, modeling uncertainty via entropy but not tracking assertion classes on edges. AutoRD [21] applies GPT-4 to rare disease literature; RECAP-KG [22] handles messy GP notes; FHIR-Ontop-OMOP [23] generates virtual KGs from OMOP data. All share a common limitation: assertion status is not propagated into the final graph.

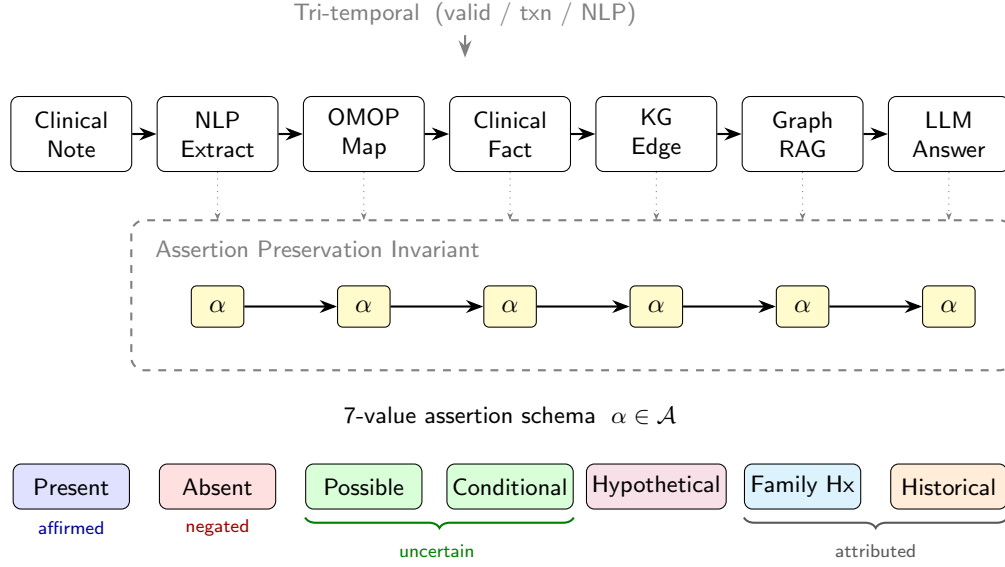


Figure 1: The EpiKG pipeline. The assertion label α (highlighted) is preserved as a first-class attribute from NLP extraction through OMOP mapping, fact construction, KG materialization, and assertion-aware GraphRAG.

Assertion detection. NegEx [24] introduced trigger-based negation detection; ConText [25] extended it with temporality and experienter. The i2b2/VA shared task [1] formalized a six-class taxonomy; Gul et al. [2] fine-tuned LLMs to 0.962 accuracy. All treat assertion detection as a terminal annotation task: labels are not carried into knowledge graphs or retrieval systems.

Assertion detection is necessary but not sufficient: labels must also be *temporally situated*—the same condition may be present in one encounter and absent in the next.

Temporal knowledge graphs. MedTKG [10] constructs temporal KGs with time-stamped snapshots (event time only); Graphiti [11] implements bitemporal edges [26] but lacks clinical ontology alignment; TEO [13] and TCL [14] formalize Allen’s relations [12] as annotation ontologies or modal operators rather than KG edge attributes. Uncertainty quantification in biomedical KGs [27] has not been connected to assertion taxonomies. Two structural gaps emerge: an *epistemic propagation gap* (assertion-detecting systems do not persist labels into KGs) and a *temporal integration gap* (temporal formalisms operate as annotation layers, not as KG edge attributes participating in retrieval). EpiKG closes both by carrying assertion and temporal metadata as first-class properties through every pipeline stage.

3 System Design

3.1 Overview

EpiKG transforms unstructured clinical narratives into an assertion-aware, temporally grounded knowledge graph exposed through graph-augmented retrieval. The pipeline comprises six stages: (1) document ingestion, (2) NLP extraction, (3) OMOP concept normalization [3], (4) clinical fact construction with epistemic metadata, (5) KG materialization with shared concept nodes, and (6) assertion-aware retrieval. The key invariant is that *epistemic assertion status* is a first-class attribute at every stage, not an NLP annotation discarded after extraction (Figure 1). Clinical concepts are stored once as shared global nodes; patient-specific semantics (assertion status, temporality, confidence) reside on edges, enabling cohort queries via edge scans.

3.2 Epistemic Assertion Schema

Clinical notes abound with qualified statements—“*no evidence of pneumonia*,” “*possible CHF*,” “*mother had breast cancer*”—yet standard representations discard these qualifiers. OMOP excludes

negated conditions entirely [3]; FHIR limits assertion metadata to conditions. EpiKG defines a seven-value assertion taxonomy:

$$\alpha \in \{\text{Pres., Abs., Poss., Cond., Hypo., Fam.Hx., Hist.}\} \quad (1)$$

This extends the i2b2 (Informatics for Integrating Biology and the Bedside) six-class taxonomy [1] by separating HISTORICAL from FAMILY_HISTORY, a clinically important distinction that prior work conflates (definitions in Appendix Q). A rule-based classifier (101 scope-aware trigger patterns) assigns α with an associated confidence score; we chose rule-based classification for deterministic behavior and interpretability, though fine-tuned LLM classifiers [2] could be substituted. Intrinsic classifier evaluation is deferred; the end-to-end ClinicalBench ablation (C3→C4) serves as a functional test of assertion propagation. The label is carried through every stage: from ExtractedMention through ensemble merging, ClinicalFact records, KG edge properties, and retrieval scoring. To our knowledge, no prior system [2, 28] maintains this end-to-end invariant.

3.3 Tri-Temporal Knowledge Graph Model

Clinical events unfold across multiple time dimensions that prior systems model incompletely [10, 11]. Bi-temporal databases [26] distinguish valid time from transaction time; EpiKG adds a third dimension (NLP-asserted temporality) and stores all three on every KG edge:

Definition 1 (Tri-Temporal Edge). *An edge $e = (s, p, o, \alpha, \tau_v, \tau_t, \tau_a, r, c)$ where:*

- s, o are source and target nodes; p is the predicate (one of 24 edge types);
- $\alpha \in \mathcal{A}$ is the assertion label (Eq. 1);
- $\tau_v = (\text{event_date}, \text{valid_from}, \text{valid_to})$ is the **valid time** interval—when the relationship held in the real world;
- $\tau_t = (\text{recorded_at}, \text{doc_date}, \text{created_at})$ is the **transaction time**—when it was recorded;
- $\tau_a \in \{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$ is the **NLP-asserted temporality**;
- $r \in \mathcal{R}$ is an Allen interval algebra relation [12]; $c \in [0, 1]$ is temporal confidence.

$\mathcal{R}$ comprises seven of Allen’s 13 canonical relations [12] (merging symmetric pairs like Before/Meets) plus CONCURRENT and UNKNOWN (nine total; full mapping in Appendix R). Unlike TEO [13] and TCL [14], which use Allen’s relations as annotation-ontology classes or modal operators, EpiKG stores r and c directly on materialized edges, enabling efficient temporal filtering during traversal.

3.4 Assertion-Aware Graph-Augmented Retrieval

Given a clinical question q and patient π , the retrieval pipeline: (1) extracts concepts from q via NLP + OMOP enrichment; (2) traverses 2–3 hops via bidirectional breadth-first search (BFS) over patient KG edges and OMOP vocabulary relationships (20M+ edges) via PostgreSQL common table expressions (CTEs) (Appendix K), pruning edges with $c < 0.3$; (3) groups edges into four temporal views (event timeline, current state, historical, conflicts); (4) retrieves matching guideline sections; (5) scores edges by:

$$\text{score}(e) = c(e) + \mathbb{I}[\text{type}(e) \in Q_{\text{rel}}] \cdot 0.2 + \mathbb{I}[\tau_a(e) = \text{CURRENT}] \cdot 0.1 \quad (2)$$

where the bonuses for question-relevant edge types (0.2) and current temporality (0.1) were set by manual tuning on a development set of 20 questions. The surviving subgraph is serialized into structured text preserving assertion labels (e.g., “ABSENT: pneumonia”); and (6) composes graph evidence, temporal context, guidelines, and source documents into a single prompt.

3.5 Formal Epistemic Preservation

We formalize the epistemic invariant maintained by the EpiKG pipeline, building on the ontological principle that aboutness must be preserved across transformations [29]. The *epistemic state* of a clinical mention m is the tuple $e(m) = (c, \alpha, \xi, \tau)$: OMOP concept, 7-value assertion label, experiencer ($\{\text{PATIENT}, \text{FAMILY}\}$), and temporality ($\{\text{CURRENT}, \text{PAST}, \text{FUTURE}\}$); see Table 6 (Appendix) for notation summary. A pipeline *epistemically preserves* m if the output epistemic state equals the extraction state at every stage. We formalize (Appendix B) that an assertion-blind pipeline collapses the assertion distribution to a degenerate point mass, incurring entropy loss

Table 1: ClinicalBench conditions. C1–C4g form the ablation ladder; C6 and C7 are bookend baselines.

ID	Condition	Retrieval	Assertion	Temporal
C1	LLM Alone	None	None	None
C2	+ Vanilla RAG	TF-IDF doc chunks	None	None
C3	+ KG-RAG (no assertions)	Graph + Doc	None	None
C4	+ Epistemic KG-RAG	Graph + Doc	Full (7-class)	Tri-temporal
C4g	+ Intent-Aware KG-RAG	Graph + Doc (type-specific)	Full (7-class)	Tri-temporal
C6	Long Context	All notes	None	None
C7	Deterministic KG	KG lookup (no LLM)	Full	Tri-temporal

$\Delta H = H(A_c^\pi) \geq 0$ [30], and that assertion-faithful accuracy is bounded by $1 - f_{np}(c)$, where f_{np} is the fraction of non-present assertions. The empirical consequences are measured via category-stratified accuracy in Section 5.

3.6 Intent-Aware Retrieval (C4g)

The multi-hop BFS described above treats all questions uniformly, yet different clinical question types require fundamentally different graph operations. A *change* question requires cross-admission set differencing; a *current-state* question needs filtering to the most recent valid edges; a *historical* question must recover resolved conditions.

Intent classifier. A lightweight rule-based classifier maps each question to one of four intents—CHANGE, CURRENT_STATE, HISTORICAL, or DEFAULT—using question metadata with keyword-pattern fallback (Algorithm 1, Appendix S).

Change retrieval. For $\iota = \text{CHANGE}$, edges are partitioned by admission (`hadm_id`). The module computes set differences across admission pairs (k, k') —concepts *added* $(C_{k'} \setminus C_k)$, *removed* $(C_k \setminus C_{k'})$, and *continued* $(C_k \cap C_{k'})$ —and returns structured evidence.

Current-state and historical retrieval. For CURRENT_STATE, the module filters to edges with $\tau_a = \text{CURRENT}$ or open-ended validity, deduplicates by concept, and emits “NOT FOUND” for missing concepts. For HISTORICAL, it selects edges with $\tau_a = \text{PAST}$ and augments with admission-based inference (concepts in earlier admissions but absent from the latest are labeled “resolved”). Each intent triggers a type-specific prompt template that structures evidence according to question semantics (Appendix U).

4 Benchmark Design

Existing clinical QA benchmarks evaluate factual recall (MedQA [31]) or agent task completion [32] but do not test handling of epistemic qualifiers or reasoning over longitudinal records of varying complexity. We introduce two complementary benchmarks built on MIMIC-IV [33].

4.1 ClinicalBench: Assertion-Sensitive Clinical QA

ClinicalBench comprises 400 gold-standard questions over 43 MIMIC-IV patients in two tasks: **Task A** (200 questions, negation-aware retrieval) and **Task B** (200 questions, temporal reasoning). Questions span 9 categories: *negation*, *conditional*, *uncertainty*, *family_history*, *sequence*, *current_state*, *duration*, *historical*, and *change*. Gold-standard answers were created by a single physician (board-certified emergency medicine) through manual chart review against source MIMIC-IV documents. No inter-annotator agreement was computed for gold-standard creation; the preregistered two-physician adjudication (Section 4.4) addresses this limitation. A pilot adjudication ($n = 5$, Appendix N) found 3/5 gold standards fully correct, motivating the full validation. Four ablation conditions progressively add system components (Table 1). Two bookend baselines—C6 (all notes concatenated) and C7 (deterministic KG lookup, no LLM)—bracket the design space.

Endpoints. The **pre-registered primary endpoint** is the hard longitudinal subset (*change* + *current_state* + *historical*, $n = 130$): C4g vs. C1 aggregate accuracy. The **secondary endpoint** is full 400-question ClinicalBench, C4g vs. C1. **Diagnostic endpoints** are per-category C1→C4g accuracy deltas and the C3→C4g transition isolating epistemic annotation lift.

What each transition isolates. C1→C2 isolates the *document retrieval effect*: C2 adds TF-IDF chunk retrieval over clinical notes, analogous to existing medical RAG systems (DoctorRAG, MedRAG).

199 C2→C3 isolates the *graph structure effect*: both retrieve documents, but C3 replaces TF-IDF with
 200 graph traversal—testing whether structured retrieval outperforms vanilla retrieval. C3→C4g iso-
 201 lates the *epistemic annotation effect*: both retrieve from the same graph, but C4g carries full 7-class
 202 assertions, tri-temporal metadata, and intent-aware routing. C4g vs. C6 isolates the *structured-vs-*
 203 *unstructured retrieval effect* on the same questions.

204 4.2 SliceBench: Complexity-Stratified Longitudinal QA

205 SliceBench tests the hypothesis that KG-augmented retrieval provides increasing benefit as pa-
 206 tient record complexity grows. We select 6 MIMIC-IV patients in three tiers: **Tier A** (2 patients,
 207 1–2 encounters), **Tier B** (2 patients, 5–10), and **Tier C** (2 patients, 15+). Each patient receives 24
 208 questions (including hard longitudinal categories: cross-encounter medication timelines, problem
 209 list reconciliation, causal chain tracing), yielding 144 total. Five conditions (B0–B4) form a mono-
 210 tone context progression from parametric-only to full system (Appendix J). The critical comparison
 211 is B2→B3: both have the same documents, but B3 adds structured KG context with assertion meta-
 212 data and temporal relations.

213 4.3 Evaluation Protocol

214 **ClinicalBench** uses a *deterministic keyword evaluator* (v2) that performs exact word-boundary
 215 matching of gold-standard assertion and temporal keywords against model answers (keyword lists
 216 in Appendix P). The v2 evaluator includes an *abstention detection gate*: when a model responds
 217 that information is unavailable in the notes (e.g., “cannot determine,” “not mentioned”) rather than
 218 making a clinical claim, the answer is scored as incorrect regardless of keyword matches. This pre-
 219 vents false positives where refusal language incidentally matches gold-standard patterns. Sequence
 220 and change categories additionally require domain-specific keywords (ordering terms and change
 221 terms, respectively). This evaluator is fully reproducible (no stochastic LLM judge) and is used for
 222 all ClinicalBench results. The primary answering model is Claude Opus 4.6; cross-model validation
 223 uses MedGemma 27B and GPT-OSS 20B under the same evaluator.

224 **SliceBench** uses an *LLM-as-judge* protocol with separated answering and judging models to pre-
 225 vent self-evaluation bias [34]: Claude Sonnet 4.5 generates answers and Claude Opus 4.6 judges
 226 correctness.

227 **Statistical reporting.** We report BCa bootstrap 95% CIs ($n = 2,000$, seed 42) [35] on both bench-
 228 marks. Question-level resampling is the primary analysis; patient-level cluster bootstrap (resampling
 229 the 43 patients with replacement, including all their questions) is reported as a sensitivity analysis
 230 to account for within-patient correlation (Appendix Table 15). McNemar’s test [36] provides a com-
 231plementary non-parametric paired comparison. ClinicalBench CIs appear on primary deltas (overall
 232 C4g–C1 and hard longitudinal subset); SliceBench CIs appear on all conditions (Table 5). A safety
 233 score with asymmetric weighting ($w = 2.0$ for false-positive assertion errors) captures clinical risk
 234 (Appendix D).

235 4.4 Physician Validation Protocol

236 To validate automated scoring, we preregister a two-physician blinded adjudication. Two board-
 237 certified emergency medicine physicians independently score a stratified sample of 168 questions
 238 under two conditions (C1 vs. C4g), blinded to condition assignment, using the same binary cor-
 239 rectness rubric as the keyword evaluator. A third physician adjudicates disagreements; inter-rater
 240 reliability is reported as Cohen’s κ . The protocol (Appendix O) was finalized before examining
 241 primary results. Results are pending.

242 5 Results

243 We evaluate EpiKG through two benchmarks: ClinicalBench tests assertion-sensitive and lon-
 244 gitudinal reasoning, while SliceBench measures longitudinal QA across patient complexity tiers.
 245 ClinicalBench uses a deterministic keyword evaluator for reproducibility; SliceBench uses LLM-as-
 246 judge evaluation (Section 4.3). Claude Opus 4.6 is the primary answering model; MedGemma 27B
 247 and GPT-OSS 20B¹ provide cross-model validation. Both benchmarks report bias-corrected and

¹GPT-OSS 20B is an open-source general-purpose model run locally via Ollama (4-bit quantized). Its identity is anonymized during review; the reproducibility package includes raw predictions for verification.

Table 2: ClinicalBench primary results (400 questions, Claude Opus 4.6, keyword evaluator v2 with abstention detection). Ablation ladder (C1–C4g) progressively adds components; C6 and C7 are bookend baselines. Sig: * denotes BCa CI excluding zero. Best in **bold**.

	Condition	Accuracy	Δ vs C1
C1	LLM Alone	21.8%	—
C2	+ Vanilla RAG	50.5%	+28.7 pp *
C3	+ KG-RAG (no assertions)	50.0%	+28.2 pp *
C4g	+ Intent-Aware KG-RAG	69.0%	+47.2 pp *
C6	Long Context (all notes)	39.0%	+17.2 pp
C7	Deterministic KG (no LLM)	3.8%	−18.0 pp

248 accelerated (BCa) bootstrap 95% CIs ($n = 2,000$, seed 42) [35], resampled at the question level.

249 5.1 ClinicalBench: Primary Ablation

250 Intent-aware KG-RAG (C4g) reaches 69.0% (+47.2 pp vs. C1; 95% BCa CI: [+40.9 pp, +52.8 pp];
 251 patient-level cluster bootstrap: [+40.2 pp, +52.5 pp]; McNemar $\chi^2 = 157.1$, $p < 10^{-6}$). Without
 252 retrieval, the model mostly abstains or produces vague responses that fail keyword matching; with
 253 structured retrieval, it produces substantive correct answers. The bookend baselines bracket the
 254 design space: C6 (all notes concatenated) scores 39.0%, outperforming C1 (+17.2 pp) because it
 255 provides clinical text, but catastrophically fails on historical (0%) and sequence (0%); C7 (deter-
 256 ministic KG, no LLM) achieves 3.8%—only negation detection works (13.6%) via stored assertion
 257 metadata—confirming that an LLM is essential for reasoning over graph evidence.

258 **Ablation decomposition: retrieval vs. graph structure vs. epistemic annotations.** C2 (vanilla
 259 TF-IDF document retrieval) and C3 (graph-augmented retrieval without assertions) score nearly
 260 identically overall—50.5% and 50.0%, respectively—demonstrating that graph structure alone does
 261 not outperform simple document retrieval in aggregate. However, per-category patterns diverge:
 262 C3 excels on current state (54% vs. 32%) where graph structure helps identify active conditions,
 263 while C2 outperforms C3 on change (33% vs. 17%) where raw text explicitly mentions medication
 264 changes. Neither approaches C4g (69.0%), which adds epistemic annotations and intent-aware rout-
 265 ing (+19.0 pp over C3; 95% BCa CI: [+13.8 pp, +24.0 pp]). The change category remains the smok-
 266 ing gun: C2 reaches 33.3%, C3 only 16.7%, but C4g achieves 100%—because change detection
 267 requires comparing assertion states across admissions, which neither vanilla retrieval nor assertion-
 268 stripped graphs can provide. C2 also serves as the closest analog to existing medical RAG systems
 269 (DoctorRAG, MedRAG), which use document retrieval without assertion-aware graph structure; the
 270 +18.5 pp C2→C4g gap quantifies the value of epistemic KG-RAG over vanilla approaches.

271 **Evaluator bias.** The keyword evaluator’s abstention gate disproportionately penalizes C1, where
 272 the model correctly recognizes it lacks retrieval context and abstains. This may inflate the apparent
 273 C1→C4g gap; the preregistered physician adjudication (Section 5.5) will quantify any systematic
 274 evaluator bias.

275 5.2 Category×Condition Interaction

276 Table 3 and Figure 2 reveal a strong *category×condition interaction*: intent-aware KG-RAG im-
 277 proves all 9 categories over C1, with every delta exceeding +35 pp. Per-category results are diag-
 278 nostic and exploratory; no multiplicity correction is applied. The preregistered primary endpoint is
 279 the hard longitudinal subset (change + current state + historical). Under the v2 evaluator with ab-
 280 stention detection, C1 drops sharply because the model without retrieval context frequently abstains
 281 (“insufficient information in the notes”) rather than attempting an answer.

282 **Where intent-aware routing excels.** The largest gains occur on categories requiring cross-
 283 admission synthesis: change (+93 pp), sequence (+55 pp), historical (+54 pp), family history
 284 (+53 pp), conditional (+45 pp), and current state (+44 pp). On the hard longitudinal subset (change
 285 + current state + historical, $n = 130$), C4g outperforms C1 by +59.2 pp (95% BCa CI: [+49.2 pp,
 286 +67.7 pp]; patient-level: [+48.6 pp, +69.4 pp]). All nine category improvements exceed +35 pp (Ap-
 287 pendix Figure 4). C6 fails catastrophically on historical (0%) and sequence (0%) despite Opus’s

Table 3: ClinicalBench per-category accuracy (%) across conditions (Claude Opus 4.6, evaluator v2). C2 adds vanilla document retrieval; C3 adds graph-structured retrieval without assertions; C4g adds epistemic annotations and intent-aware routing. Best per row in **bold**.

Category	n	C1	C2	C3	C4g	C4g−C1
Negation	110	45.5	70.9	65.5	80.9	↑ 35.5 pp
Conditional	20	0.0	35.0	30.0	45.0	↑ 45.0 pp
Uncertainty	40	12.5	35.0	37.5	50.0	↑ 37.5 pp
Family hist.	30	3.3	43.3	43.3	56.7	↑ 53.3 pp
Sequence	40	10.0	50.0	57.5	65.0	↑ 55.0 pp
Current st.	50	26.0	32.0	54.0	70.0	↑ 44.0 pp
Duration	30	30.0	66.7	60.0	66.7	↑ 36.7 pp
Historical	50	6.0	48.0	42.0	60.0	↑ 54.0 pp
Change	30	6.7	33.3	16.7	100.0	↑ 93.3 pp
Overall	400	21.8	50.5	50.0	69.0	↑ 47.2 pp

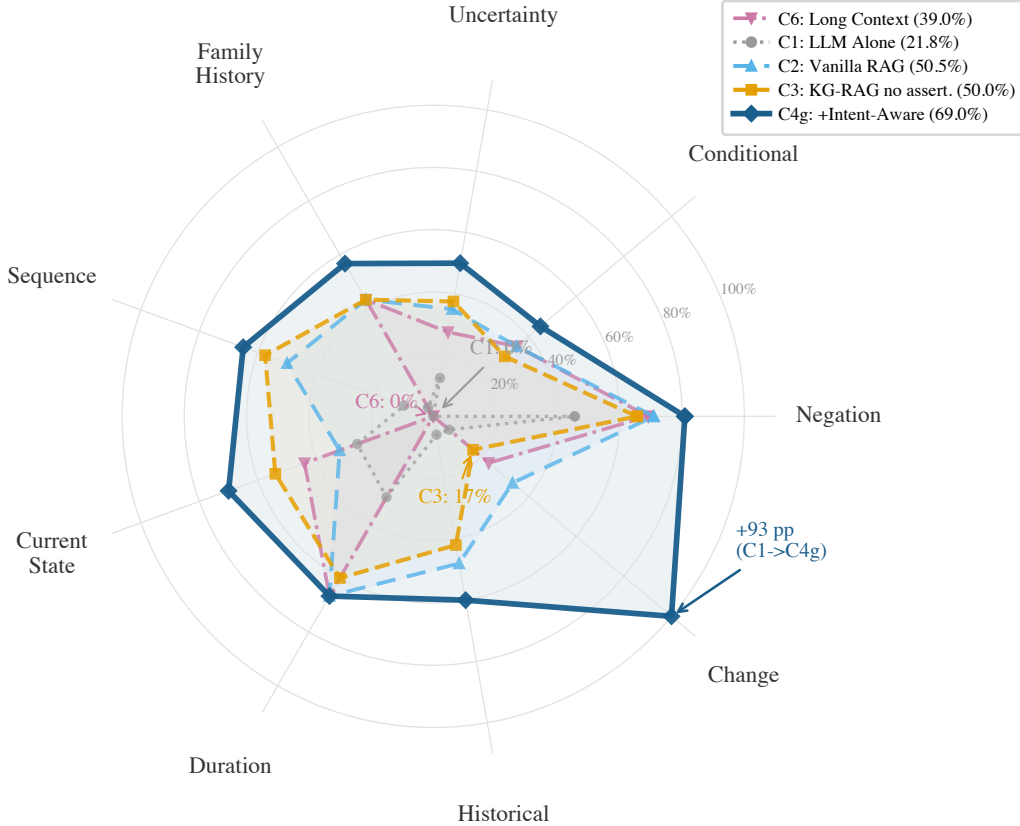


Figure 2: ClinicalBench per-category accuracy across five conditions (C6, C1, C2, C3, C4g; Claude Opus 4.6, evaluator v2). C6 (pink dash-dot) collapses on historical and sequence (both 0%); C1 (gray dotted) falls below 50% on 7 of 9 categories; C2 (light blue) and C3 (orange dashed) overlap at ~50% overall but diverge per category; C4g (dark solid) adds epistemic annotations, improving all 9 categories over C1. The C3→C4g gap on change (16.7%→100%) highlights the critical role of assertion metadata.

200K window, confirming that long context cannot substitute for structured retrieval on these categories.

5.3 Cross-Model Validation

All three models benefit substantially from KG-RAG (Table 4), with the magnitude consistent with an inverse relationship to parametric medical knowledge. Opus gains +47.2pp, GPT-OSS

Table 4: ClinicalBench cross-model results (400 questions, keyword evaluator v2). All three models benefit from KG-RAG; deltas vary by model.

	Model	C1	C4g	Δ
Commercial	Claude Opus 4.6	21.8%	69.0%	+47.2 pp
Open	GPT-OSS 20B	20.5%	58.0%	+37.5 pp
Medical	MedGemma 27B	26.2%	57.8%	+31.5 pp

Table 5: SliceBench overall results (144 questions, Claude Sonnet 4.5, judged by Opus 4.6). Sig: * denotes CI excluding zero.

	Condition	Score	95% CI	Δ vs B0
B0	LLM Alone	49.9%	[43.6%, 56.1%]	—
B1	Latest Note	73.4%	[67.8%, 78.7%]	+23.5 pp
B2	All Notes RAG	84.7%	[80.2%, 88.8%]	+34.8 pp *
B3	KG-RAG	86.9%	[82.8%, 90.8%]	+37.0 pp *
B4	Full System	87.6%	[83.7%, 91.2%]	+37.8 pp *

gains +37.5 pp, and MedGemma—the only model fine-tuned on medical corpora—gains +31.5 pp. Notably, MedGemma starts with the highest C1 baseline (26.2%) yet achieves the lowest C4g accuracy (57.8%), while Opus starts at 21.8% but reaches 69.0% with retrieval.

All three models improve on all 9 categories under C4g (per-category cross-model breakdowns in Appendix Table 8). Intent-aware KG-RAG thus functions as a *medical knowledge compensator*: the structured epistemic context partially substitutes for parametric medical knowledge, providing the largest benefit to general-purpose models that need it most.

5.4 SliceBench

The incremental KG layer (B2→B3) contributes +2.2 pp overall (CI: [-1.5 pp, +5.9 pp]), not reaching aggregate significance, but tier-stratified estimates show a complexity-dependent trend: +0.6 pp (Tier A), +1.0 pp (Tier B), +5.0 pp (Tier C, 15+ encounters), consistent with KG context being most valuable when document volume overwhelms chunk-level retrieval (Appendix L).

5.5 Physician Adjudication

A two-physician blinded adjudication ($n = 168$ ClinicalBench questions $\times$ 2 conditions, Cohen’s κ for inter-rater reliability) is preregistered to validate automated scoring; the protocol and blinded scoring sheets (Appendix O) were designed before examining results. *Results are pending.*

6 Discussion and Limitations

Why long context underperforms. Brute-force long context (C6) scores 39.0%, outperforming C1 (+17.2 pp) because it at least provides clinical text for the model to draw on, but it still falls far short of C4g (69.0%) and catastrophically fails on historical (0%) and sequence (0%). C6 concatenates all notes chronologically without retrieval or selection; more sophisticated long-context strategies (hierarchical summarization, temporal segmentation) might perform better, but C6’s category-level failures—consistent with the “lost in the middle” phenomenon [37]—demonstrate that raw context dumping cannot substitute for structured retrieval. Historical questions require distinguishing resolved from active conditions across admissions; sequence questions require strict temporal ordering—both are implicit in raw text but explicit in the epistemic KG. C6 handles change detection modestly (23%) because the relevant information appears explicitly in the text, but KG-RAG achieves 100% (+77 pp).

Model-dependent KG benefit. Across the three models tested, KG-RAG benefit is consistent with an inverse relationship to parametric medical knowledge (Table 4). The model with the lowest C1 baseline (Opus, 21.8%) achieves the highest C4g accuracy (69.0%), while the medically fine-tuned model (MedGemma) starts highest (26.2%) but gains least under C4g. Intent-aware KG-RAG thus functions as a *medical knowledge compensator*: the structured epistemic context partially substitutes for parametric knowledge, providing the largest benefit to models that need it most. This has

practical implications for RAG system design: the same retrieval pipeline may yield different gains depending on the downstream model, and single-model evaluation risks overstating or understating value.

When KG context matters. The complexity-dependent SliceBench effect (Tier C: +5.0 pp vs. Tier A: +0.6 pp) suggests structured epistemic context is most valuable when: (a) document volume overwhelms chunk-level retrieval, (b) questions require cross-encounter synthesis where the same concept appears with different assertion states, and (c) assertion status evolves across encounters. The C3 ablation provides the clearest evidence: structured retrieval without assertions lifts overall accuracy from 21.8% to 50.0% (+28.2 pp), but change detection reaches only 16.7%—because detecting what *changed* between admissions requires comparing assertion states, which C3 strips. Adding epistemic annotations (C3→C4g) lifts change to 100% (+83.3 pp), confirming that *assertion metadata*, not mere graph structure, is the key factor for change detection.

Uncertainty remains weakest. Uncertainty improves from 12.5% to 50.0% (+37.5 pp), but absolute accuracy remains low. Possible and suspected conditions are linguistically subtle (“consider statin,” “may have CHF”) and require pragmatic inference beyond keyword matching.

Limitations. Sample size is modest: ClinicalBench covers 43 patients (400 questions) and SliceBench 6 patients (144 questions), both from MIMIC-IV [33]. The B2→B3 CI crosses zero ([−1.5 pp, +5.9 pp]); tier-level estimates ($n=2$ patients per tier) are hypothesis-generating only. Model-dependent effects mean headline gains are specific to Opus; gains differ across models (Table 4). The keyword evaluator may miss semantically correct answers; physician adjudication (Section 5.5) will address this. This is an ablation study; direct comparisons with medical RAG systems [5, 6, 15] were not performed because these target factoid QA rather than patient-level longitudinal reasoning. C2 (vanilla TF-IDF document retrieval, 50.5%) serves as the closest proxy for these systems on our task, demonstrating that vanilla retrieval reaches roughly the same aggregate accuracy as graph-augmented retrieval without assertions (C3, 50.0%) but both fall ~19 pp short of epistemic KG-RAG (C4g, 69.0%). Patient-level cluster bootstrap CIs (Appendix Table 15) are slightly wider than question-level CIs but all primary endpoints remain significant; the 43-patient sample nonetheless limits generalization. MedGemma run-to-run variance (± 10 pp) arises from GPU non-determinism in 4-bit quantized inference. Over-reliance on KG-driven summaries risks anchoring bias if NLP extraction introduces systematic errors; single-site evaluation (US academic center) leaves generalization unvalidated (Appendix V).

Conclusion. The epistemic propagation gap is a real and measurable phenomenon. EpiKG addresses it through end-to-end epistemic preservation combined with intent-aware retrieval that matches graph traversal to question semantics. The C3 ablation decomposes the gain: structured retrieval contributes the majority (+28 pp), while epistemic annotations are decisive for change detection (+83 pp from C3 to C4g). The benefit is both category-specific (all 9 improve, with the largest gains on longitudinal reasoning) and model-dependent (knowledge-compensator pattern across three models). Neither more context nor structured data alone suffices—C6 and C7 bracket the design space—confirming that the category×condition interaction, not the aggregate, is the right lens for clinical KG-RAG evaluation.

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470 A Notation Summary

Symbol	Meaning
$\alpha \in \mathcal{A}$	Assertion label (7 values, Eq. 1)
τ_v	Valid time (event date, valid from/to)
τ_t	Transaction time (recorded at, doc date, created at)
τ_a	NLP-asserted temporality (Current / Past / Future)
$r \in \mathcal{R}$	Temporal relation (9 values, Table 18)
c	Confidence score $\in [0, 1]$
ξ	Experiencer (Patient / Family)
$e(m)$	Epistemic state tuple (c, α, ξ, τ)
q, π	Clinical question, patient
ι	Question intent (Change / Current_State / Historical / Default)
$f_{np}(c)$	Fraction of non-present assertions for concept c

Table 6: Key notation used throughout the paper.

471 B Formal Epistemic Preservation

472 We formalize the epistemic invariant maintained by the EpiKG pipeline, building on the principle
473 that aboutness—the relationship between information artifacts and the entities they represent—must
474 be preserved across transformations [29].

475 **Definition 2** (Epistemic State). *For a clinical mention m , the epistemic state is the tuple $e(m) =$
476 (c, α, ξ, τ) , where c is the OMOP concept identifier, $\alpha \in \mathcal{A}$ is the 7-value assertion label (Eq. 1), $\xi \in$
477 $\{\text{PATIENT, FAMILY}\}$ is the experiencer, and $\tau \in \{\text{CURRENT, PAST, FUTURE}\}$ is the temporality. A
478 pipeline P epistemically preserves mention m if $P(e(m)) = e(m)$; that is, the epistemic state at the
479 output of every pipeline stage is identical to the state at extraction.*

Proposition 3 (Assertion Entropy Loss). *For concept c in patient π ’s record, let A_c^π denote the
assertion distribution with empirical frequencies $p(\alpha_i) = n_i / \sum_j n_j$ over the $|\mathcal{A}|$ assertion classes.
The assertion entropy is*

$$H(A_c^\pi) = - \sum_i p(\alpha_i) \log p(\alpha_i) . \quad (3)$$

480 An assertion-blind pipeline collapses all mentions to $\alpha = \text{PRESENT}$, yielding a degenerate distribu-
481 tion with $H = 0$. The information loss is $\Delta H = H(A_c^\pi) \geq 0$ [30], with $\Delta H > 0$ strictly whenever
482 any mention carries a non-present assertion.

483 *Proof.* Under the assertion-blind mapping $\phi : \alpha_i \mapsto \text{PRESENT}$ for all i , the output distribution
484 assigns probability 1 to PRESENT and 0 to all other classes, so $H(\phi(A_c^\pi)) = 0$. By non-negativity
485 of Shannon entropy, $\Delta H = H(A_c^\pi) - 0 = H(A_c^\pi) \geq 0$, with equality iff all mentions already carry
486 $\alpha = \text{PRESENT}$. $\square$

487 **Corollary 4** (Faithfulness Bound). *Let $f_{np}(c)$ denote the fraction of mentions of concept c carrying*
488 *a non-present assertion ($\alpha \neq \text{PRESENT}$). Without assertion labels, the maximum assertion-faithful*
489 *accuracy for any downstream task conditioned on c is bounded by $1 - f_{np}(c)$. In clinical records*
490 *where negated and uncertain mentions are prevalent—e.g., “no pneumonia,” “possible CHF”—this*
491 *bound can be substantially below 1.*

492 *Proof.* Without assertion labels, any predictor must assign a single assertion class to all mentions
493 of c . Choosing PRESENT (the majority class in clinical text) yields accuracy $1 - f_{np}(c)$; the $f_{np}(c)$
494 non-present mentions are necessarily misclassified. $\square$

495 C Gap Analysis

496 Partial marks indicate limited coverage: GFM-RAG, MedRAG, and KARE build population-level
497 or hierarchical KGs (not per-patient longitudinal graphs); KARE uses KG-augmented retrieval but

Table 7: Capability comparison across representative systems. ✓ = supported, ○ = partial, ✗ = not supported.

System	Patient KG	OMOP mapping	Assertion (7-class)	Tri-temporal	Assert-aware RAG	Experienter	Allen's algebra	Multi-hop (≥ 3)
DoctorRAG [15]	✗	✗	✗	✗	✗	✗	✗	✗
GFM-RAG [5]	○	✗	✗	✗	✗	✗	✗	✓
MedRAG [16]	○	✗	✗	✗	✗	✗	✗	○
KARE [6]	○	✗	✗	✗	○	✗	✗	✓
Multi-LLM KG [20]	✗	✓	○	✗	✗	✗	✗	✗
MedTKG [10]	✓	✗	✗	○	✗	✗	✗	✓
Graphiti [11]	✗	✗	✗	○	✗	✗	✗	✓
MediGRAF [18]	✓	✗	✗	✗	✗	✗	✗	✓
EpiKG (ours)	✓	✓	✓	✓	✓	✓	✓	○

without assertion awareness; Multi-LLM KG models uncertainty via entropy but not the full assertion taxonomy; MedTKG and Graphiti use temporal annotations but not tri-temporal models. MediGRAF is the closest patient-level competitor, constructing per-patient medical graphs from clinical notes, but it does not preserve assertion status or temporal relations as edge attributes. EpiKG’s partial mark on multi-hop traversal (≥ 3 hops) reflects a deliberate architectural trade-off: PostgreSQL CTE-based traversal provides ACID compliance but degrades beyond 2 hops (Appendix K).

D Safety Score

The safety score $S = 1 - \frac{1}{N} \sum_{i=1}^N w_i \cdot \mathbb{I}[\hat{y}_i \neq y_i]$ applies $w_i = 2.0$ for false-positive assertion errors (reporting a negated or absent condition as present) and $w_i = 1.0$ otherwise, capturing asymmetric clinical risk where acting on a falsely affirmed condition is more dangerous than missing a present one. We treat $w = 2.0$ as a reasonable default; sensitivity to this choice is a direction for future work.

E Cross-Model Per-Category Results

Table 8: ClinicalBench per-category accuracy (%) by model under C1 and C4g (intent-aware KG-RAG, evaluator v2). All three models improve on all 9 categories.

Category	n	Claude Opus 4.6			MedGemma 27B			GPT-OSS 20B		
		C1	C4g	Δ	C1	C4g	Δ	C1	C4g	Δ
Negation	110	45.5	80.9	↑35.5 pp	57.3	76.4	↑19.1 pp	46.4	80.0	↑33.6 pp
Conditional	20	0.0	45.0	↑45.0 pp	15.0	35.0	↑20.0 pp	0.0	15.0	↑15.0 pp
Uncertainty	40	12.5	50.0	↑37.5 pp	7.5	32.5	↑25.0 pp	5.0	30.0	↑25.0 pp
Family hist.	30	3.3	56.7	↑53.3 pp	3.3	40.0	↑36.7 pp	3.3	40.0	↑36.7 pp
Sequence	40	10.0	65.0	↑55.0 pp	7.5	47.5	↑40.0 pp	12.5	42.5	↑30.0 pp
Current st.	50	26.0	70.0	↑44.0 pp	32.0	58.0	↑26.0 pp	36.0	56.0	↑20.0 pp
Duration	30	30.0	66.7	↑36.7 pp	53.3	70.0	↑16.7 pp	16.7	70.0	↑53.3 pp
Historical	50	6.0	60.0	↑54.0 pp	0.0	46.0	↑46.0 pp	0.0	60.0	↑60.0 pp
Change	30	6.7	100.0	↑93.3 pp	0.0	76.7	↑76.7 pp	0.0	70.0	↑70.0 pp
Overall	400	21.8	69.0	↑47.2 pp	26.2	57.8	↑31.5 pp	20.5	58.0	↑37.5 pp

F MedGemma Full Per-Category Results

Category	n	C7	C1	C4g
Negation	110	13.6	57.3	76.4
Conditional	20	0.0	15.0	35.0
Uncertainty	40	0.0	7.5	32.5
Family History	30	0.0	3.3	40.0
Sequence	40	0.0	7.5	47.5
Current State	50	0.0	32.0	58.0
Duration	30	0.0	53.3	70.0
Historical	50	0.0	0.0	46.0
Change	30	0.0	0.0	76.7
Overall	400	3.8	26.2	57.8

Table 9: MedGemma 27B ClinicalBench per-category accuracy (% , keyword evaluator v2) for three conditions. MedGemma gains +32 pp overall under C4g, with all 9 categories improving. Strongest improvements on change (+77 pp) and historical (+46 pp).

512 G Experienter Attribute Propagation Ablation

513 The experienter attribute distinguishes patient conditions from family member conditions;
514 without it, the graph conflates the two, causing family history misattribution.

Table 10: Impact of experienter attribute propagation on Opus C4 (400 questions, prior evaluator run). Categories with no change omitted.

Category	Pre-fix	Post-fix	Δ
Family history	63.3%	73.3%	+10.0 pp
Uncertainty	50.0%	55.0%	+5.0 pp
Historical	34.0%	38.0%	+4.0 pp
Current state	46.0%	48.0%	+2.0 pp
Overall C4	68.2%	70.2%	+2.0 pp

515 The fix improved family history by +10.0 pp with zero regressions on guard categories (negation,
516 conditional, duration, sequence all unchanged), confirming that the experienter attribute is load-
517 bearing for assertion-sensitive reasoning.

518 H SliceBench Figures

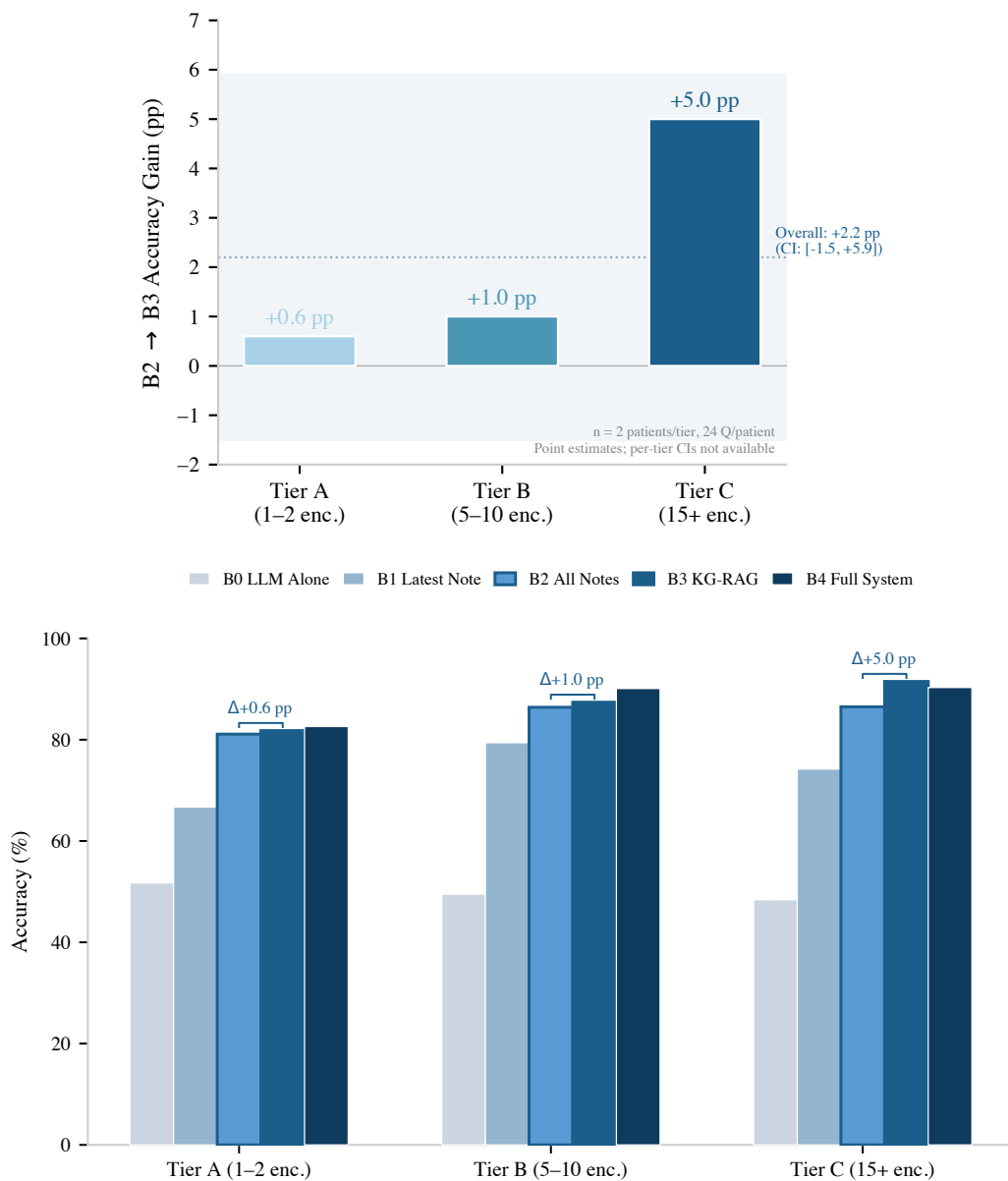


Figure 3: SliceBench results ($n = 2$ patients per tier, 24 questions each; point estimates without per-tier CIs). **Top:** The KG layer’s incremental benefit (B2→B3) shows a directional, complexity-dependent trend: +0.6 pp (Tier A) to +5.0 pp (Tier C). **Bottom:** Full condition progression by tier. The overall B2→B3 delta is +2.2 pp (CI: [-1.5 pp, +5.9 pp]).

519 I Per-Category Delta Chart

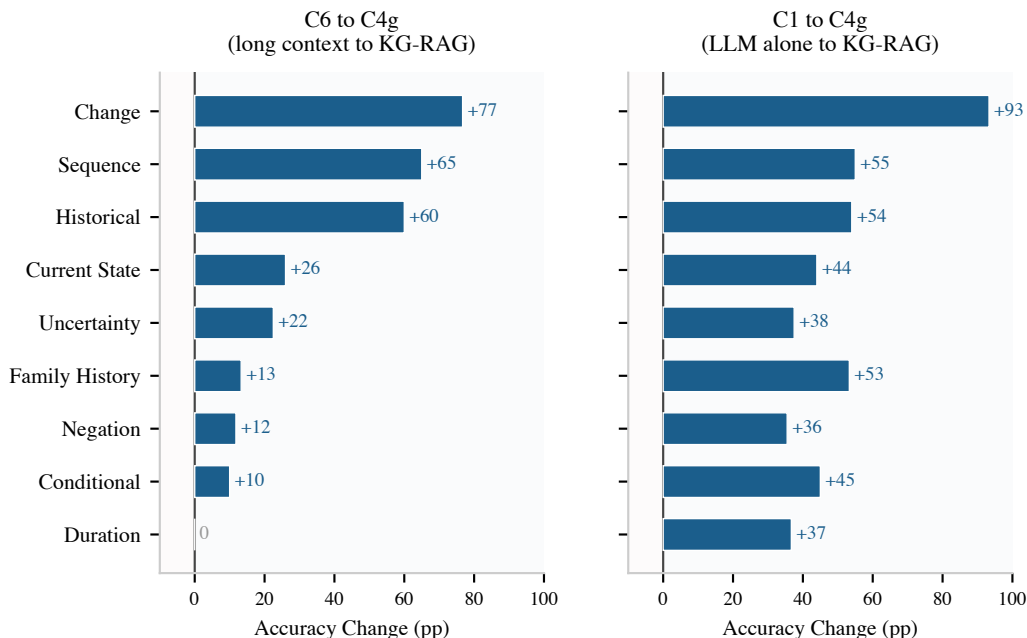


Figure 4: Per-category accuracy change (Claude Opus 4.6, evaluator v2) for two comparisons. **Left:** C6→C4g (structured vs. unstructured retrieval): KG-RAG outperforms long context on all 9 categories, with the largest gains on sequence (+65 pp) and historical (+60 pp). **Right:** C1→C4g (LLM alone to KG-RAG): all 9 categories improve—change (+93 pp), sequence (+55 pp), historical (+54 pp).

520 J SliceBench Conditions

Table 11: SliceBench conditions. B0–B4 form a monotone context progression.

ID	Condition	Context Provided to LLM
B0	LLM Alone	Question only (parametric knowledge)
B1	Latest Note	Most recent clinical note
B2	All Notes RAG	All notes, retrieved by relevance
B3	KG-RAG	Knowledge graph context + all notes
B4	Full System	KG-RAG + guidelines + calculators

521 K System Implementation Details

522 **LLM baseline.** On MedQA-USMLE (965 questions), Claude Opus 4.5 achieves 81.6% accu-
523 racy (5.1 pp below GPT-4 at 86.7% and Med-PaLM 2 at 86.5% [32]), establishing that the Claude
524 model family—Sonnet 4.5 (SliceBench answerer) and Opus 4.6 (ClinicalBench primary, SliceBench
525 judge)—provides a reasonable LLM baseline (Table 12).

Model	Accuracy	Step 1	N
EpiKG (LLM alone)	81.6%	79.9%	965
GPT-4 (2023)	86.7%	—	1,273
Med-PaLM 2 (2023)	86.5%	—	1,273
Claude 3 Opus (2024)	78.0%	—	1,273

Table 12: MedQA-USMLE results. Our LLM-alone baseline is competitive.

526 **KG scale and traversal latency.** The deployed system processes 145 documents across 85 pa-
527 tients, materializing 3,100 KG nodes and 8,803 edges. Graph traversal latency is sub-millisecond
528 at this scale: 0.57 ms (1-hop), 0.75 ms (2-hop). The full system integrates 201 clinical calculators,

1,202 guideline sections, 20M+ OMOP vocabulary relationships, 101 assertion trigger patterns, and 24 edge types across 13 node types.

Multi-hop traversal. On the DR.KNOWS diagnostic reasoning benchmark [38], which measures KG traversal accuracy using PostgreSQL-backed clinical data, EpiKG achieves 0.420 overall (50% at 1-hop, 25% at 2-hop, 0% at 3-hop). Multi-hop degradation reflects a deliberate architectural trade-off: PostgreSQL CTE-based traversal provides ACID compliance but scales poorly beyond 2 hops.

L SliceBench Tier-Stratified Results

Table 13: SliceBench results stratified by patient complexity tier. B2→B3 Δ shows the incremental KG contribution.

Tier	Encounters	Score (%)					
		B0	B1	B2	B3	B4	B2→B3 Δ
A	1–2 notes	51.7	66.7	81.1	81.8	82.6	+0.6 pp
B	5–10 notes	49.5	79.4	86.4	87.4	90.1	+1.0 pp
C	15+ notes	48.4	74.2	86.5	91.5	90.3	+5.0 pp

M ClinicalBench Full Per-Category Results (Opus)

Category	n	C7	C6	C1	C2	C3	C4g
Negation	110	13.6	69.1	45.5	70.9	65.5	80.9
Conditional	20	0.0	35.0	0.0	35.0	30.0	45.0
Uncertainty	40	0.0	27.5	12.5	35.0	37.5	50.0
Family History	30	0.0	43.3	3.3	43.3	43.3	56.7
Sequence	40	0.0	0.0	10.0	50.0	57.5	65.0
Current State	50	0.0	44.0	26.0	32.0	54.0	70.0
Duration	30	0.0	66.7	30.0	66.7	60.0	66.7
Historical	50	0.0	0.0	6.0	48.0	42.0	60.0
Change	30	0.0	23.3	6.7	33.3	16.7	100.0
Overall	400	3.8	39.0	21.8	50.5	50.0	69.0

Table 14: Complete ClinicalBench per-category accuracy (%), Claude Opus 4.6, keyword evaluator v2) for all conditions. C7 is model-independent; C2 adds vanilla document retrieval; C3 adds graph-structured retrieval without assertion metadata. C2 and C3 are nearly identical overall ($\sim 50\%$) but diverge per category. All 9 categories improve from C1 to C4g; the largest gains occur on change (+93 pp), sequence (+55 pp), and historical (+54 pp). C3→C4g on change (+83 pp) isolates the epistemic annotation contribution.

N Human Validation Pilot

To assess LLM-as-judge reliability, we conducted a pilot validation study with a single board-certified physician who scored system outputs on a 4-point scale without knowledge of condition labels. The pilot ($n = 5$) revealed overly strict automated scoring on uncertainty and sequence questions. Gold standards were fully correct in 3/5 cases; model answers were partially correct or better in 3/5 cases where the auto-scorer marked them incorrect. A full two-physician blinded adjudication is preregistered (Section 5.5, Appendix O).

O Physician Validation Protocol

Design. Two board-certified emergency medicine physicians independently score ClinicalBench answers under two conditions (C1 and C4g) using the same binary correctness rubric as the deterministic keyword evaluator. Reviewers are blinded to condition assignment. The sample comprises 168 ClinicalBench questions stratified across all 9 categories (oversampling categories at highest evaluator error risk), with both conditions scored per question (336 total scored pairs). Blinded scoring sheets with randomized condition labels (A/B) are included in the supplementary materials.

Inter-rater agreement. Cohen’s κ is computed between the two reviewers on the full 336 scored pairs. A third board-certified physician adjudicates all disagreements.

Endpoints. (1) Human–keyword evaluator concordance rate (fraction of automated scores confirmed by physicians). (2) Human-adjudicated C4g vs. C1 accuracy delta. (3) Category-level human accuracy.

Disclosure. Both primary reviewers are co-authors. We report κ to demonstrate independent judgment and disclose the relationship in the main text.

Status. This protocol was designed before examining primary results. *Results are pending.*

P Reproducibility Details

Models. ClinicalBench primary ablation: Claude Opus 4.6 (claude-opus-4-20250514, keyword evaluator v2). Cross-model: MedGemma 27B (gemma3:27b, 4-bit GGUF) and GPT-OSS 20B (4-bit GGUF) via Ollama; same evaluator. SliceBench: Claude Sonnet 4.5 (claude-sonnet-4-5-20250929) answers, Opus 4.6 judges. Temperature 0 throughout; 4-bit quantization introduces GPU non-determinism (± 10 pp run-to-run for MedGemma), controlled via within-run paired comparisons.

Evaluation. The deterministic keyword evaluator performs exact word-boundary matching (regex `\bKEYWORD\b`) against gold-standard assertion and temporal keywords. Per-category keyword sets: negation (“no,” “denies,” “absent,” “negative”), uncertainty (“possible,” “suspected,” “may,” “likely,” “consider”), temporal (“before,” “after,” “during,” “changed,” “new”), plus category-specific terms. Evidence preambles (echoed graph context) are stripped before matching.

Retrieval. C2’s vanilla RAG uses TF-IDF over chunked clinical notes (512-token chunks, 64-token overlap), retrieving the top-5 chunks by cosine similarity. This is the closest analog to existing medical RAG systems (DoctorRAG, MedRAG) on patient-level longitudinal data. ClinicalBench conditions map to SliceBench: C1 $\approx$ B0, C2 $\approx$ B2, C4 $\approx$ B3, C5 $\approx$ B4.

Bootstrap. $n = 2,000$ resamples, seed 42, BCa method [35]. We report both question-level and patient-level (cluster) bootstrap CIs; the latter resample the 43 patients with replacement, including all their questions per draw, to account for within-patient correlation. Patient-level CIs are slightly wider but all primary endpoints remain significant (Table 15).

McNemar’s test. We supplement bootstrap CIs with McNemar’s test for paired nominal data, comparing discordant pairs (C1 wrong/C4g right vs. C1 right/C4g wrong). All three pairwise comparisons are significant: C1 vs. C4g ($\chi^2 = 157.1$, $p < 10^{-6}$; discordant: 18 vs. 207), C1 vs. C3 ($\chi^2 = 73.4$, $p < 10^{-6}$; 29 vs. 142), and C3 vs. C4g ($\chi^2 = 49.3$, $p < 10^{-11}$; 19 vs. 95).

Table 15: Statistical summary: question-level and patient-level (cluster) BCa bootstrap 95% CIs for primary endpoints. Both levels exclude zero for all comparisons.

Comparison	Δ	Question CI	Patient CI	McNemar p
C4g–C1 (overall)	+47.2 pp	[+40.9 pp, +52.8 pp]	[+40.2 pp, +52.5 pp]	$< 10^{-6}$
C3–C1 (retrieval)	+28.2 pp	[+22.3 pp, +34.0 pp]	[+23.1 pp, +33.8 pp]	$< 10^{-6}$
C4g–C3 (epistemic)	+19.0 pp	[+13.8 pp, +24.0 pp]	[+14.6 pp, +23.3 pp]	$< 10^{-11}$
Hard longitudinal	+59.2 pp	[+49.2 pp, +67.7 pp]	[+48.6 pp, +69.4 pp]	—

Data. De-identified MIMIC-IV clinical notes accessed under a PhysioNet Credentialed Health Data Use Agreement (v1.5.0).

Compute. ClinicalBench Opus C1/C3/C4g: ~ 22 min each (API); Opus C6: ~ 3.5 h (API); C7: < 1 min (no LLM); MedGemma C1–C4g: ~ 3.6 h (single GPU); MedGemma C6: ~ 1.5 h; SliceBench: ~ 2 h (API).

Evaluator evolution. The keyword evaluator underwent three versions: v0 (substring matching), v1 (word-boundary matching + evidence preamble stripping), and v2 (+ abstention detection gate + domain-specific keyword requirements for sequence and change). The v1 evaluator awarded false

positives when models responded with “insufficient information in the notes” because negation and temporal keywords in the refusal text matched gold-standard patterns. The v2 evaluator adds an abstention detection layer: answers matching abstention patterns (e.g., “cannot determine,” “not mentioned,” “insufficient evidence”) are scored as incorrect unless they contain clinical claim patterns (e.g., “patient does not,” “denies”). Additionally, v2 requires sequence answers to contain ordering keywords (“first,” “then,” “before”) and change answers to contain change keywords (“added,” “removed,” “discontinued”), preventing term-overlap-only false positives. The duration category’s minimum-0.5 score floor for matching duration keywords was removed. The v1→v2 transition reduced C1 accuracy (from ~50% to 21.8% for Opus) by correctly classifying abstention responses as incorrect; C4g accuracy decreased modestly (from ~76% to 69.0%) because the model abstains less frequently when retrieval context is provided. All main-text numbers use v2.

Reproducibility package. A standalone reproducibility package is included as supplementary material (epikg-benchmark/). It contains all 400 ClinicalBench questions with gold answers, raw model predictions for all condition×model combinations (Opus C1/C3/C4g/C6, MedGemma C1/C4g, GPT-OSS C1/C4g), and the deterministic keyword evaluator v2 with abstention detection. Running the evaluator requires only Python 3.10+ with no external dependencies. All accuracy numbers in Tables 2, 3, and 4 are independently reproducible from this package. MIMIC-IV patient identifiers are included so that PhysioNet-credentialed reviewers can trace predictions back to source clinical notes.

Results provenance. Table 16 maps each reported result to its source checkpoint file. MedGemma C4g has 399 predictions (1 timeout); the missing question is scored as incorrect ($n = 400$).

Table 16: Results provenance: checkpoint files for each condition×model combination. All scored with keyword evaluator v2.

Condition	Model	Checkpoint file	n
C1	Opus 4.6	opus/C1_llm_alone.json	400
C2	Opus 4.6	opus/C2_vanilla_rag.json	392*
C3	Opus 4.6	opus/C3_kg_rag.json	400
C4g	Opus 4.6	opus/C4g_intent_aware.json	400
C6	Opus 4.6	opus/C6_long_context.json	400
C7	—	opus/C7_deterministic.json	400
C1	MedGemma 27B	medgemma/C1_llm_alone.json	400
C4g	MedGemma 27B	medgemma/C4g_intent_aware.json	399*
C1	GPT-OSS 20B	gptoss/C1_llm_alone.json	400
C4g	GPT-OSS 20B	gptoss/C4g_intent_aware.json	400

*C2: 8 sequence questions returned empty (retrieval timeout); MedGemma C4g: 1 timeout.

All missing answers scored as incorrect for $n = 400$ denominator.

Q Assertion Category Definitions

Assertion	Definition	Example
PRESENT	Condition currently affirmed	“has diabetes”
ABSENT	Condition explicitly negated	“denies chest pain”
POSSIBLE	Condition suspected, not confirmed	“possible pneumonia”
CONDITIONAL	Contingent on specific circumstances	“if febrile, start antibiotics”
HYPOTHETICAL	Discussed as a scenario	“would need dialysis if...”
FAMILY_HISTORY	Attributed to a family member	“mother had breast cancer”
HISTORICAL	Previously true, not necessarily current	“former smoker”

Table 17: The 7-value assertion taxonomy, extending i2b2 by separating HISTORICAL from FAMILY_HISTORY.

R Temporal Relation Mapping

The nine temporal relations $\mathcal{R}$ used on KG edges are derived from Allen’s 13 canonical interval relations [12] by merging symmetric pairs.

$\mathcal{R}$ value	Allen source(s)	Meaning
BEFORE	Before, Meets	A ends before B starts
AFTER	After, Met-by	A starts after B ends
DURING	During	A entirely within B
CONTAINS	Contains	A entirely contains B
OVERLAPS	Overlaps, Overlapped-by	A and B partially overlap
STARTS	Starts, Started-by	A and B share start time
FINISHES	Finishes, Finished-by	A and B share end time
CONCURRENT	Equals	A and B have same interval
UNKNOWN	—	Relation undetermined

Table 18: Mapping from Allen’s 13 interval relations to the 9 temporal relation values stored on KG edges.

S Intent-Aware Routing Algorithm

The C4g classifier is rule-based, mapping questions to one of four intents using keyword patterns (e.g., “changed”, “new since” trigger CHANGE; “currently”, “active problem” trigger CURRENT_STATE). Algorithm 1 details the full retrieval procedure.

Algorithm 1: Intent-Aware Retrieval (C4g)

Input: Question q , patient π
Output: Structured evidence E

```

1  $\mathcal{C} \leftarrow \text{EXTRACTCONCEPTS}(q)$ ; // NLP + OMOP enrichment
2  $\iota \leftarrow \text{CLASSIFYINTENT}(q)$ ; //  $\in \{\text{CHANGE}, \text{CURRST}, \text{HIST}, \text{DEFAULT}\}$ 
3 if  $\iota = \text{CHANGE}$  then
4   Partition edges by admission:  $\mathcal{E}_k \leftarrow \{e \mid \text{hadm\_id}(e) = k\}$ ;
5   foreach admission pair  $(k, k')$  with  $k < k'$  do
6      $\mathcal{A} \leftarrow \mathcal{C}_{k'} \setminus \mathcal{C}_k$ ;  $\mathcal{R} \leftarrow \mathcal{C}_k \setminus \mathcal{C}_{k'}$ ;  $\mathcal{S} \leftarrow \mathcal{C}_k \cap \mathcal{C}_{k'}$ ;
7    $E_g \leftarrow \text{FORMATCHANGE}(\mathcal{A}, \mathcal{R}, \mathcal{S})$ ;
8 else if  $\iota = \text{CURRST}$  then
9    $E_g \leftarrow \text{FILTEREDGES}(\pi, \mathcal{C}, \tau_a = \text{CURRENT} \vee \text{open validity})$ ;
10  Deduplicate by concept; emit “NOT FOUND” for missing  $c \in \mathcal{C}$ ;
11 else if  $\iota = \text{HIST}$  then
12   $E_g \leftarrow \text{FILTEREDGES}(\pi, \mathcal{C}, \tau_a = \text{PAST})$ ;
13  Augment: concepts in earlier admissions but absent from latest  $\rightarrow$  “resolved”;
14 else
15   $E_g \leftarrow \text{BIDIRECTIONALBFS}(\pi, \mathcal{C}, \text{hops} = 2-3, c_{\min} = 0.3)$ ;
16  $E_d \leftarrow \text{RETRIEVEDOCUMENTS}(\pi, \mathcal{C})$ ;
17 return  $\text{COMPOSE}(E_g, E_d, \text{template}(\iota))$ ;

```

T Bookend and Diagnostic Conditions

C5 (Full System). C5 extends C4 with guideline retrieval (1,202 sections) and clinical calculators (201, e.g., $\text{CHA}_2\text{DS}_2\text{-VASC}$, MELD). In a prior evaluator run, C5 achieved 68.2% overall, below the prior-run C1 and C4, likely because non-relevant components dilute the context window.

C6 (Long Context). All patient documents are concatenated chronologically and presented to the LLM. For Opus (200K token window), all notes fit; for MedGemma (8K window), later documents are truncated. C6 achieves 39.0% (Opus) overall—better than C1 (21.8%) but far below

628 C4g (69.0%). Opus catastrophically fails on historical (0%) and sequence (0%) under C6, confirm-
629 ing that brute-force long context cannot substitute for structured retrieval on categories requiring
630 cross-encounter synthesis.

631 **C7 (Deterministic KG).** KG edges matching query concepts are returned directly without LLM
632 reasoning. C7 achieves 3.8% overall (only negation at 13.6% via stored assertion metadata), con-
633 firming that structured data without an LLM reasoner cannot answer clinical questions.

634 U Illustrative Retrieval Example

635 To illustrate why intent-aware routing matters, consider a change question: “*What medications*
636 *changed between this patient’s first and second admissions?*”

637 **C1 (LLM alone).** The model sees only the current note, which mentions metoprolol, atorvastatin,
638 and “discontinued lisinopril.” Without prior admission records, it cannot determine what was *added*
639 vs. *continued*, and may hallucinate prior medication lists.

640 **C4g (intent-aware KG-RAG).** The intent classifier routes to CHANGE retrieval, which partitions
641 KG edges by `hadm_id` and computes set differences:

- 642 • **Added** (admission 2 only): atorvastatin 40mg [PRESENT]
- 643 • **Removed** (admission 1 only): lisinopril 10mg [PRESENT→ABSENT]
- 644 • **Continued**: metoprolol 25mg [PRESENT, both admissions]

645 The assertion labels (PRESENT, ABSENT) disambiguate: “discontinued lisinopril” is not a current
646 medication but a historical one whose status changed—exactly the distinction the epistemic schema
647 preserves.

648 V Ethics Statement

649 This work uses de-identified clinical data from MIMIC-IV [33] under a PhysioNet Credentialed
650 Health Data Use Agreement. No patient re-identification was attempted. The system is designed for
651 clinical decision *support*, not autonomous clinical decision-making.

652 NeurIPS Paper Checklist

653 1. Claims

654 Question: Do the main claims made in the abstract and introduction accurately reflect the
655 paper’s contributions and scope?

656 Answer: **Yes.**

657 Justification: The abstract and introduction state four specific contributions (Section 1), each
658 supported by empirical evidence in Sections 5.1–5.4. SliceBench claims include bootstrap
659 CIs; ClinicalBench claims use within-run paired comparisons (Section 4.3). Claims are
660 scoped with “to our knowledge” qualifiers where appropriate.

661 2. Limitations

662 Question: Does the paper discuss the limitations of the work performed by the authors?

663 Answer: **Yes.**

664 Justification: Section 6 explicitly discusses limitations including small sample sizes (6 patients
665 for SliceBench, 2 per tier), single-site data (MIMIC-IV), the B2→B3 confidence interval
666 crossing zero, model-dependent effects, and pending physician adjudication.

667 3. Theory assumptions and proofs

668 Question: For each theoretical result, does the paper provide the full set of assumptions and a
669 complete (correct) proof?

670 Answer: **Yes.**

671 Justification: Proposition 3 (assertion entropy loss) and Corollary 4 (faithfulness bound) in
672 Appendix B state all assumptions explicitly. These are information-theoretic characterizations
673 with all steps included.

674 4. Experimental result reproducibility

675 Question: Does the paper fully disclose all the information needed to reproduce the main
676 experimental results of the paper to the extent that it affects the main claims and/or conclusions
677 of the paper?

678 Answer: **Yes.**

679 Justification: Appendix P reports exact model versions (MedGemma 27B, Claude Sonnet 4.5,
680 Claude Opus 4.6), temperature settings (0), bootstrap parameters ($n = 2,000$, seed 42, BCa
681 method), and data source (MIMIC-IV v1.5.0 under PhysioNet DUA). Section 4 describes
682 question generation and evaluation protocols.

683 5. Open access to data and code

684 Question: Does the paper provide open access to the data and code, with sufficient instructions
685 to faithfully reproduce the main experimental results?

686 Answer: **Yes.**

687 Justification: A standalone reproducibility package is included as supplementary material
688 (`epikg-benchmark/`) containing all 400 questions, raw predictions for six condition $\times$ model
689 combinations, and the deterministic keyword evaluator. All main-text accuracy numbers are
690 independently reproducible. MIMIC-IV source notes require separate PhysioNet credentialed
691 access. Full system code will be released upon acceptance.

692 6. Experimental setting/details

693 Question: Does the paper specify all the training and test details necessary to understand the
694 results?

695 Answer: **Yes.**

696 Justification: Section 4 specifies benchmark design, question counts, condition definitions,
697 patient selection criteria, and evaluation methodology. Appendix P provides model versions
698 and computational requirements.

699 7. Experiment statistical significance

700 Question: Does the paper report error bars suitably and correctly?

701 Answer: **Yes.**

702 Justification: Table 5 reports BCa bootstrap 95% CIs for all conditions. Per-tier results (Fig-
703 ure 3) are clearly labeled as point estimates without per-tier CIs due to small tier-level sample
704 sizes ($n = 2$ patients per tier). The overall B2 $\rightarrow$ B3 CI crossing zero is explicitly noted.

705 8. Experiments compute resources

706 Question: For each experiment, does the paper provide sufficient information on the computer
707 resources needed to reproduce the experiments?

708 Answer: **Yes.**

709 Justification: Appendix P reports computational requirements: ClinicalBench Opus ≈ 22 min
710 per condition (API), MedGemma ≈ 3.6 hours (single GPU), SliceBench ≈ 2 hours (API).

711	9. Code of ethics
712	Question: Does the research conducted in the paper conform, in every respect, with the
713	NeurIPS Code of Ethics?
714	Answer: Yes.
715	Justification: The study uses de-identified clinical data under an approved data use agreement
716	(Appendix V). No patient re-identification was attempted. The system is designed for clinical
717	decision <i>support</i> , not autonomous decision-making.
718	10. Broader impacts
719	Question: Does the paper discuss both potential positive societal impacts and negative societal
720	impacts of the work?
721	Answer: Yes.
722	Justification: Section 6 discusses the potential for improved clinical decision support (positive)
723	and risks of over-reliance on automated systems, context dilution, and the need for physician
724	oversight (negative). Appendix V addresses data ethics.
725	11. Safeguards
726	Question: Does the paper describe safeguards that have been put in place for responsible
727	release of data or models?
728	Answer: Yes.
729	Justification: The system operates on de-identified data under PhysioNet DUA. It is designed
730	as a decision support tool requiring physician oversight. Code release will include usage
731	guidelines emphasizing the system is not validated for autonomous clinical use.
732	12. Licenses for existing assets
733	Question: Are the creators or original owners of assets used in the paper properly credited and
734	are the license and terms of use explicitly mentioned and properly respected?
735	Answer: Yes.
736	Justification: MIMIC-IV is cited [33] and used under PhysioNet Credentialed Health Data
737	Use Agreement. All benchmark frameworks and models are cited. OMOP CDM is an open
738	standard.
739	13. New assets
740	Question: Are new assets introduced in the paper well documented and is the documentation
741	provided alongside the assets?
742	Answer: Yes.
743	Justification: ClinicalBench and SliceBench are new benchmarks fully documented in Sec-
744	tion 4, with question generation procedures, category definitions (Appendix Q), and evalua-
745	tion protocols. Templates and scoring scripts will accompany the code release.
746	14. Crowdsourcing and research with human subjects
747	Question: For crowdsourcing experiments and research with human subjects, does the paper
748	include the full text of instructions given to participants and screenshots?
749	Answer: N/A.

750 Justification: No crowdsourcing was used. Physician adjudication (Appendix N) involves
751 expert review by co-authors, not human subjects research.

752 **15. IRB approvals or equivalent**

753 Question: Does the paper describe potential risks and does it include the IRB approval number
754 if applicable?

755 Answer: **N/A.**

756 Justification: MIMIC-IV is a de-identified public dataset; use under PhysioNet DUA does not
757 require separate IRB approval per PhysioNet policy.

758 **16. Declaration of LLM usage**

759 Question: Does the paper describe the usage of LLMs if it is an important, original, or non-
760 standard component of the core methods?

761 Answer: **Yes.**

762 Justification: LLMs are core experimental components, not authoring aids. Claude Opus 4.6
763 is the primary answering model for ClinicalBench (Section 5.1); Claude Sonnet 4.5 answers
764 SliceBench questions while Opus 4.6 judges (Section 5.4); MedGemma 27B provides cross-
765 model validation (Section 5.3). Model versions, temperatures, and inference details are in
766 Appendix P.